

Strassen's Matrix Multiplication

Base case:

if $n = 2$

→ (1)

$$\begin{aligned}C_{11} &= a_{11} * b_{11} + a_{12} * b_{21} \\C_{12} &= a_{11} * b_{12} + a_{12} * b_{22} \\C_{21} &= a_{21} * b_{11} + a_{22} * b_{21} \\C_{22} &= a_{21} * b_{12} + a_{22} * b_{22}\end{aligned}$$

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}_{2 \times 2} \times B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}_{2 \times 2} = C = \begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix}$$

if $n = 1$

↳ 8 multiplications

$$C = [a_{11} * b_{11}]$$

$$A = [a_{11}] \quad b = [b_{11}]$$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}$$

A

$$B = \begin{bmatrix} b_{11} & b_{12} & b_{13} & b_{14} \\ b_{21} & b_{22} & b_{23} & b_{24} \\ b_{31} & b_{32} & b_{33} & b_{34} \\ b_{41} & b_{42} & b_{43} & b_{44} \end{bmatrix}$$

B

4x4

$$C = \begin{bmatrix} A_{11} * B_{11} + A_{12} * B_{21} & A_{11} * B_{12} + A_{12} * B_{22} \\ A_{21} * B_{11} + A_{22} * B_{21} & A_{21} * B_{12} + A_{22} * B_{22} \end{bmatrix}$$

C

Algorithm $mm(A, B, n)$ { $\rightarrow T(n)$

if $(n \leq 2)$ {

use eq (1): $\rightarrow 1$

}
else {

mid = $n/2$;

$mm(A_{11}, B_{11}, n/2)$ + $mm(A_{12}, B_{21}, n/2)$;

$mm(A_{11}, B_{12}, n/2)$ + $mm(A_{12}, B_{21}, n/2)$;

$mm(A_{21}, B_{11}, n/2)$ + $mm(A_{22}, B_{21}, n/2)$;

$mm(A_{21}, B_{12}, n/2)$ + $mm(A_{22}, B_{22}, n/2)$;

}
}

$\hookrightarrow$ matrix addition

$$T(n) = 8T(n/2) + n^2$$

$$\text{So, } a=8 \quad b=2 \quad k=2 \quad p=0$$

$$\log_2 8 > 2$$

$$3 > 2$$

we use case 2

$$O(n^{\log_2 8})$$

$$O(n^3)$$

Strassen's Seven Product

↳ Majority time is taken for multiplication. So, he reduced it from 8 to 7. at the expense of more + & -.

He formulated:

$$\begin{aligned}P &= (A_{11} + A_{22})(B_{11} + B_{22}) \\Q &= (A_{21} + A_{22})B_{11} \\R &= A_{11}(B_{12} - B_{22}) \\S &= A_{22}(B_{21} - B_{11}) \\T &= (A_{11} + A_{12})B_{22} \\U &= (A_{21} - A_{11})(B_{11} + B_{12}) \\V &= (A_{12} - A_{22})(B_{21} + B_{22})\end{aligned}$$

This has 7 multiplication

$$\begin{aligned}C_{11} &= P + S - T + V \\C_{12} &= R + T \\C_{21} &= Q + S \\C_{22} &= P + R - Q + U\end{aligned}$$

Instead of using eq (i), we use abo

$$T(n) = 7T(n/2) + n^2$$

$$\text{So, } a=7 \quad b=2 \quad k=2 \quad p=0$$

$$\log_2 7 > 2$$

$$2.81 > 2$$

we use case 2

$$O(n^{\log_2 7})$$

$$O(n^{2.81})$$

